

SOBANA R S

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PROFESSIONAL SUMMARY

Motivated and detail-oriented AI & Data Science graduate with hands-on experience in Python development, Generative AI, Retrieval-Augmented Generation (RAG), and Machine Learning. Demonstrated ability to design and deploy end-to-end AI solutions across healthcare, manufacturing, and enterprise domains. Passionate about building intelligent systems using LLMs, LangChain, and modern NLP frameworks. Currently contributing as an Associate Software Engineer with a strong track record of delivering impactful projects in fast-paced environments. Seeking a challenging role in Generative AI, Python development, or applied AI engineering.

WORK EXPERIENCE

Associate Software Engineer – Python Developer | **Greywiz Private Limited** | May 2025 – Present

Domain: Python Developer – RAG Chatbot Project

Technologies: Python, LangChain, LangGraph, RAG, PostgreSQL, LLMs

- Designed and developed an end-to-end RAG (Retrieval-Augmented Generation) chatbot to answer queries using both internal documents and external knowledge sources.
- Implemented LangChain for document ingestion, embedding generation, vector storage, and intelligent retrieval pipelines.
- Utilized LangGraph to manage multi-step conversational workflows and improve response consistency across sessions.
- Integrated PostgreSQL for structured data storage and metadata management, ensuring scalable and reliable data access.

INTERNSHIPS

AI/ML Intern | **Infosys Springboard Internship 5.0 (On Stream)** | December 2024 – February 2025

Domain: NLP, Machine Learning, Generative AI

- Developed an AI-powered tool that generates role-specific interview questions for recruiters and hiring managers using Generative AI techniques.

Gen AI Intern | **Agiliz Tech, Bangalore, Karnataka** | November 2023 – December 2023

Domain: Generative AI & Prompt Engineering

- Trained on Generative AI and prompt engineering methodologies and developed a functional chatbot for a medical website.

PROJECTS & ENTERPRISE SOLUTIONS

Automatic Screw Detection System – Indo Metal Tech, Chennai (Nov 2024 – Feb 2025)

- Developed an AI-based screw counting system using OpenCV and advanced image processing techniques.
- Implemented a camera-based detection system integrated with a computer for automated counting, resulting in improved manufacturing efficiency.
- Solution was accepted and successfully deployed in the production environment.

Press Clipping Management System – Indian Railways, Madurai (Aug 2024 – Oct 2024)

- Built a web-based platform to digitize and manage daily railway news clippings for Public Relations Officers (PROs).
- Enabled categorization, annotation, and archiving of news content, significantly improving accessibility and eliminating physical storage constraints.
- Integrated a calendar-based retrieval feature for quick access to historical reports.

INNOVATIVE IMPLEMENTATIONS

Generative AI

- Code Debugger (Groq): Built an AI assistant to analyze and debug code efficiently, reducing developer effort and improving productivity.
- Sales Email Generator (Groq): Automated the generation of personalized sales emails, improving outreach efficiency for businesses.
- Research Paper Summarizer (Groq): Created a tool to condense research papers into concise summaries for researchers and students.
- Text-to-SQL Query Converter (Groq): Developed an AI system to convert natural language into SQL queries, making database querying accessible to non-technical users.
- YouTube Video Summarizer (Gemini 1.5 Flash): Built an AI tool to summarize YouTube videos, saving time for users seeking quick insights.

Machine Learning

- Supply Chain Disruption Prediction (XGBoost): Built a predictive model to forecast supply chain risks and help businesses maintain inventory balance.
- Stroke Prediction (Random Forest Classifier): Developed a risk assessment model to assist healthcare professionals in early stroke intervention.

Deep Learning

- Pneumonia Detection (TensorFlow, Keras, CNN): Built a CNN model for pneumonia diagnosis from chest X-ray images, enabling early detection.
- Skin Cancer Classification (TensorFlow, Keras, CNN): Developed an AI system for accurate and early skin cancer classification.

Natural Language Processing

- Text-to-Speech & Speech-to-Text: Created an accessibility tool for visually and hearing-impaired individuals.
- Named Entity Recognition (SpaCy): Designed an NER model to extract key entities from unstructured text, aiding information retrieval.
- Language Detection (LangDetect): Implemented a system for automatic language identification in multilingual text processing.

EDUCATION

Bachelor of Technology (B.Tech) – Artificial Intelligence & Data Science

Sethu Institute of Technology, Virudhunagar, Tamil Nadu **CGPA: 8.99 | 2021 – 2025**

Higher Secondary (Grade 12) – 89%

St. John's Matriculation Higher Secondary School, Madurai, Tamil Nadu | 2020 – 2021

Secondary (Grade 10) – 86.4%

St. John's Matriculation Higher Secondary School, Madurai, Tamil Nadu | 2018 – 2019

TECHNICAL SKILLS

Programming & Web	Python, HTML, CSS, JavaScript
ML & AI Frameworks	TensorFlow, PyTorch, Scikit-learn, OpenCV, Hugging Face
Generative AI & LLMs	LangChain, LangGraph, RAG, Groq, Gemini 1.5, Prompt Engineering
Deep Learning	CNN, Transformers, Generative AI Models
Data Science	Pandas, NumPy, Matplotlib, Seaborn
NLP & OCR	NLTK, SpaCy, Pytesseract, LangDetect
Databases	MySQL, PostgreSQL
Data Visualization	Power BI, Excel
Version Control	Git, GitHub

PUBLICATIONS

- Detectiverse – Advancing Supply Chain Efficiency with AI-Enhanced Screw Counting: Published in the Journal of Image Processing & Pattern Recognition Progress, focusing on AI-driven object detection for automated screw counting.
- Designing and Developing a Cancer Chatbot for a Website: Published in the Journal of Multimedia Technology & Recent Advancements, presenting an AI-based chatbot for assisting cancer patients with medical queries and appointment scheduling.
- IoT-Based Smart Farming: Published in Intensive Research Organisation, focusing on automated monitoring and control of agricultural activities using IoT technology.

ACHIEVEMENTS & CERTIFICATIONS

- Won 1st Prize in Project Presentation at SSM Institute of Technology.
- Secured 2nd Prize in a Hackathon organized by Oasys Cybernetics Pvt. Ltd. and Inext Labs – developed an AI-powered Automatic Recruitment Pipeline under the Generative AI theme.
- Completed UX Design certification by Google on Coursera.
- Earned certification in Advanced Data Analytics using Python on Udemy.
- Certified in Artificial Intelligence Primer by Infosys Springboard.
- Completed Machine Learning with AI using Python by NASSCOM with a 70% assessment score in the Gold category.
- Successfully completed three NPTEL courses: Cloud Computing, Internet of Things, and Joy of Computing using Python.
- Gained expertise in Python for Data Science and Machine Learning (Udemy).

PROFESSIONAL DEVELOPMENT

Pragati: Path to Future Program | Infosys, Chennai | July 2024

Participated in a comprehensive development program focused on future-ready skills and leadership, including hands-on workshops, interactive sessions, and networking with industry experts.

CONFERENCES & HACKATHONS

- Participated in the Emerging Trends in Information and Communication Technologies Conference (ETICT), gaining insights into advancements in IT and communication technologies.
- Attended the International Conference on Emerging Electrical Systems and Control (EESC), exploring innovations in electrical systems and automation.
- Developed an Accessible Video for Deaf solution at Infosys Springboard iAccelerate Women's Hackathon under the theme 'Innovation for Inclusion'.
- Competed in national-level competitions including India's G20 Competition, IEEE National Level Online Project Competition (Guru Nanak Institutions), and the National Entrepreneurship Conclave.

LANGUAGES & SOFT SKILLS

Languages: English, Tamil, Telugu | **Soft Skills:** Goal-Oriented, Time Management, Planning, Verbal & Written Communication